

DYLAN BURKEY

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EDUCATION

Georgia Institute of Technology <i>Bachelor of Science in Mechanical Engineering (Manufacturing Concentration)</i> Georgia Tech Europe (GTE) <i>Study Abroad Program</i>	Atlanta, GA Aug. 2023 – May 2027 Metz, France Jan. 2025 - May 2025
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EXPERIENCE

Amazon <i>Robotics Sortable FC Area Manager</i> Incoming Summer 2026.	May 2026 – Aug. 2026 Miami, FL
Smurfit Westrock <i>Engineering Intern</i> - Conducted root cause failure analysis on thermal and mechanical production systems, identifying design and process improvements projecting \$275K in annual savings across corrugated manufacturing equipment. - Created detailed CAD models for machine equipment and new company products to support design iteration and manufacturing documentation. - Developed a data-driven inventory optimization system (Oracle JD Edwards API integration), leveraging MATLAB and Excel for data analytics to support continuous production operations and reduce excess stock in a \$13.52M inventory by 23%. - Collaborated cross-functionally with maintenance and operations teams on equipment diagnostics, integrating lean manufacturing and 5S practices to improve machine uptime and overall equipment effectiveness.	May 2025 – Aug. 2025 Mooresville, NC
Snapping Shoals EMC <i>Automotive Intern</i> - Performed hands-on mechanical diagnostics, preventive maintenance, and repair across hydraulic, electrical, and chassis systems on a fleet of 75 utility vehicles supporting a \$12M asset base. - Supported chassis and aerial-lift system improvements including hydraulic inspection, structural integrity assessment, and component serviceability upgrades to improve assembly efficiency.	Aug. 2022 – May 2023 Covington, GA
Curious Cardinal <i>Mentor</i> Mentored K-12 students in personalized one-on-one STEM sessions, designing customized curricula and coaching students through professional development and portfolio creation.	Dec. 2024 – Present Atlanta, GA

PROJECTS

Autonomous Robot - ME 2110 CAD, Mechatronics, Prototyping and Fabrication, DFM/DFA, DFMEA - Designed and fabricated an autonomous robot with custom mechanical subsystems, integrating Arduino-based sensor and motion control to complete five tasks in under 40 seconds — placing 2nd of 70+ teams in Georgia Tech's Creative Decisions & Design competition. - Applied DFM/DFA and DFMEA through iterative prototype-to-final-design cycles using CAD, 3D printing (FDM), laser cutting, and manual machining, optimizing part reliability, tolerances, and assembly efficiency. - Produced detailed engineering drawings with GD&T specifications and led cross-functional troubleshooting of mechanical-electrical integration challenges across design iterations.	Aug. 2025 – Dec. 2025
R&D: RV-14 Aircraft (Yellow Jacket Aviation) Process & Build Sequencing, MFG & Process Coordination - Designed and validated fabrication plans and assemblies for an FAA-compliant aircraft, applying lightweight structural design principles and material selection to meet performance and manufacturability requirements. - Developed build sequencing documentation, BOMs, and kitting plans to support hardware builds, standardize assembly procedures, and reduce rework risk — coordinating across structures, systems, and logistics teams.	May 2025 – Present

INVOLVEMENT AND LEADERSHIP

Yellow Jacket Aviation <i>Research Team Lead</i> Lead of a YJA's 25-member research team, spearheading R&D of an FAA-compliant aircraft, coordinating interdisciplinary engineering efforts and collaborating with logistics and design teams for a legal, manufacturable, and cost-effective strategy.	Feb. 2023 – Present
American Society of Mechanical Engineers <i>General Body Member</i> Participate in weekly professional development events and networking sessions with top-tier engineering firms to enhance industry knowledge and career opportunities.	Sep. 2023 – Present

TECHNICAL SKILLS

CAD: Solidworks, Fusion 360, Onshape, PTC Creo
Engineering: Finite Element Analysis (FEA), Data Analytics (Matlab, Python, Excel), GD&T, Prototyping and Iteration, 3D Printing (FDM), Arduino/Microcontroller Programming, Robotic System Integration, Digital Fabrication, Controls and Actuation
Manufacturing: Lean Manufacturing (5S, Six Sigma, Kaizen, Standard Work), Process Improvement and OEE, RCA & Defect Reduction, Quality Control, Material Characterization
Certifications: Six Sigma Yellow Belt, CompTIA A+